Sankalp Rajeev

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SUMMARY

AI/ML engineer and dual M.S. candidate in Artificial Intelligence and Robotics with experience deploying production computer vision, LLM, and ML systems at Magna, Zoetis, and Mistral. Focused on model reliability, inference optimization, and production ML systems.

EDUCATION

University of Michigan - Dearborn Sep 2024 - Dec 2026
M.S., Artificial Intelligence & M.S.E., Robotics Engineering; GPA: 3.90/4.00 Dearborn, MI
Arizona State University Aug 2020 - May 2024
B.S., Computer Science; Minor: Mathematical Concepts of Engineering; GPA: 3.71/4.00 Tempe, AZ

WORK EXPERIENCE

University of Michigan - Dearborn Sep 2026 - Present
Graduate Researcher, LLM Software Repair Reliability Dearborn, MI

- Built a Docker-isolated verifier for LLM-generated software repairs across 100 SWE-bench Verified tasks, combining regression tests, static analysis, and issue-derived reproduction tests in network-disabled containers with hashed patches and 840+ automated tests.
- A pre-registered audit halted the study after detecting benchmark leakage in 8 of 50 tasks, traced to agents reading newer package copies baked into prebuilt images; added container-level masking and integrity checks before restarting the 100-task run.

Magna International May 2026 - Aug 2026
Data Science Intern, Advanced Manufacturing Innovation R&D Troy, MI

- Validated multi-camera stamped-part inspection with up to $7M/year projected labor savings, proving registration and cross-camera deduplication on 2- and 4-camera rigs using SIFT/RANSAC calibration and Hungarian matching.
- Delivered a lens-debris inspection pipeline projected to save ~$700K/year in labor, achieving 100% recall and 96% precision with high-pass filtering, CLAHE preprocessing, and PatchCore; handed the validated system off to the manufacturing plant.
- Productionized an unsupervised visual-inspection pipeline for a Magna Seating plant in Spain, reaching 100% defect recall, 95.2% precision, and ~300 ms latency on Jetson AGX Orin after benchmarking 6+ approaches and deploying YOLOX + NanoSAM + PatchCore with ONNX, TensorRT FP16, Docker, and NVIDIA Triton.

Zoetis May 2025 - Nov 2025
AI Engineering Intern Kalamazoo, MI

- Deployed 4 production LLM tools on Azure Kubernetes Service using asynchronous ETL and semantic matching, reducing regulated documentation effort by 1,000+ engineer-hours annually.
- Reduced incident misclassification by ~40% by building a real-time SQL-backed analytics workflow with Plotly dashboards and automated openpyxl Excel exports.

Mistral Solutions Pvt Ltd Jun 2024 - Aug 2024
AI/ML Engineer Intern Bangalore, India

- Reduced UAV inference latency by 25% by quantizing YOLOv8 and ByteTrack with Qualcomm SNPE on a QRB5165 edge platform; also deployed a VGG16 vehicle classifier with 96% accuracy through the same inference pipeline.
- Integrated Lightware LW20/C LiDAR and DS1820 sensors through Arduino and I2C interfaces to support real-time range and thermal telemetry for onboard perception workflows.

TECHNICAL PROJECTS

Kairos - Demand Intelligence Platform PySpark, Kafka, Airflow, PyTorch, XGBoost, Docker

- Processed 411M+ e-commerce events through PySpark, Kafka, and Airflow; built a SASRec -> Two-Tower -> XGBoost recommendation stack reaching NDCG@10 ~0.90 and AUC 0.9883.
- Built demand forecasting with 9.44% WMAPE and validated the pipeline with GitHub Actions CI and 83 automated Pytest tests.

Nexus - LLMOps Observability & Self-Healing Platform LangGraph, Kafka, FastAPI, MLflow, Docker

- Built an 11-service reliability platform that streams agent traces through Kafka and classifies hallucination, context-bleed, and plan-drift failures.
- Implemented LangGraph-based root-cause diagnosis and prompt/config repair, then re-evaluated fixes with DeepEval + MLflow in <90 s.

CodeMind - Multi-Agent Code Intelligence Platform CrewAI, LangChain, ChromaDB, Ollama, QLoRA

- Built hybrid vector + BM25 + Reciprocal Rank Fusion retrieval with 15.4 ms median latency across 42K+ chunks from 10 repositories, orchestrating agents for code exploration, security, testing, performance, and documentation.
- Fine-tuned Gemma 3 12B with QLoRA/Unsloth for automated test generation, achieving 34.7% CodeBLEU (+73.5% relative to baseline) with a 3.3x inference speedup.

TECHNICAL SKILLS

ML / Deep Learning: PyTorch, Transformers, XGBoost, scikit-learn, anomaly detection, unsupervised learning, recommender systems
Computer Vision: OpenCV, PatchCore, YOLOX/YOLOv8, SAM2/NanoSAM, object detection, feature matching, image preprocessing
LLM / AI Systems: LangGraph, LangChain, RAG, LLM evaluation, SWE-bench, Hugging Face, QLoRA, agentic workflows
MLOps / Inference: NVIDIA Triton, TensorRT, ONNX, Jetson AGX Orin, MLflow, ClearML, Docker, CI/CD, Pytest, Linux, Git, GitHub Actions
Data / Cloud: PySpark, Kafka, Airflow, Databricks, PostgreSQL, Redis, GCP, Azure Kubernetes Service
Languages: Python, C/C++, SQL, TypeScript, Java